



Photo by Gretchen Kay Stuart,
National Wildlife Federation

SUITABILITY ANALYSIS FOR CASCADE RED FOX HABITATS

Anika Nelson
August 3, 2025

TABLE OF CONTENTS

Part A – Analysis Model.....	3
Part B – Refining the Suitability Analysis.....	7
Part C – Layouts	9
Part D – Discussion.....	14
References.....	15

PART A — FIGURES

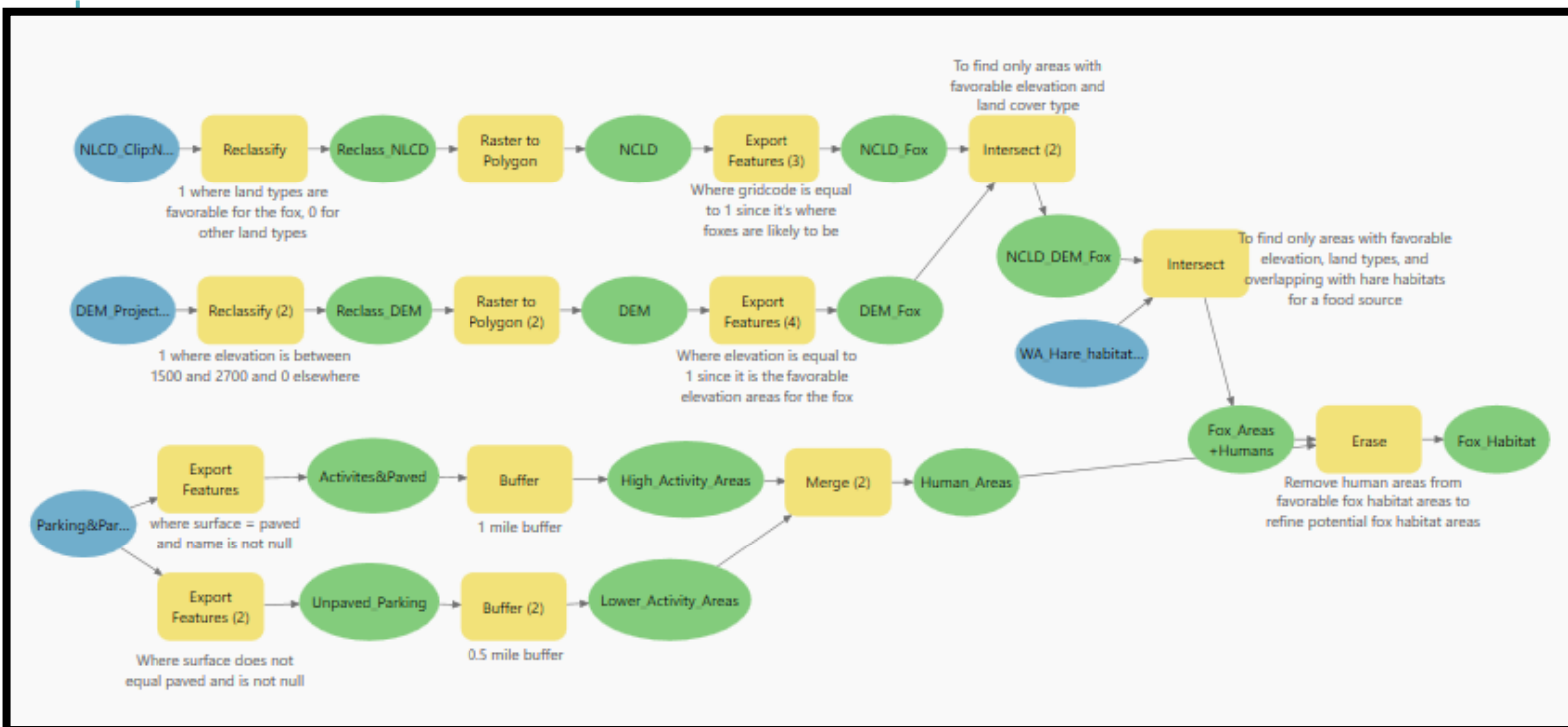


Figure 1 . Full model for determining likely habitat areas for the Cascade Red Fox

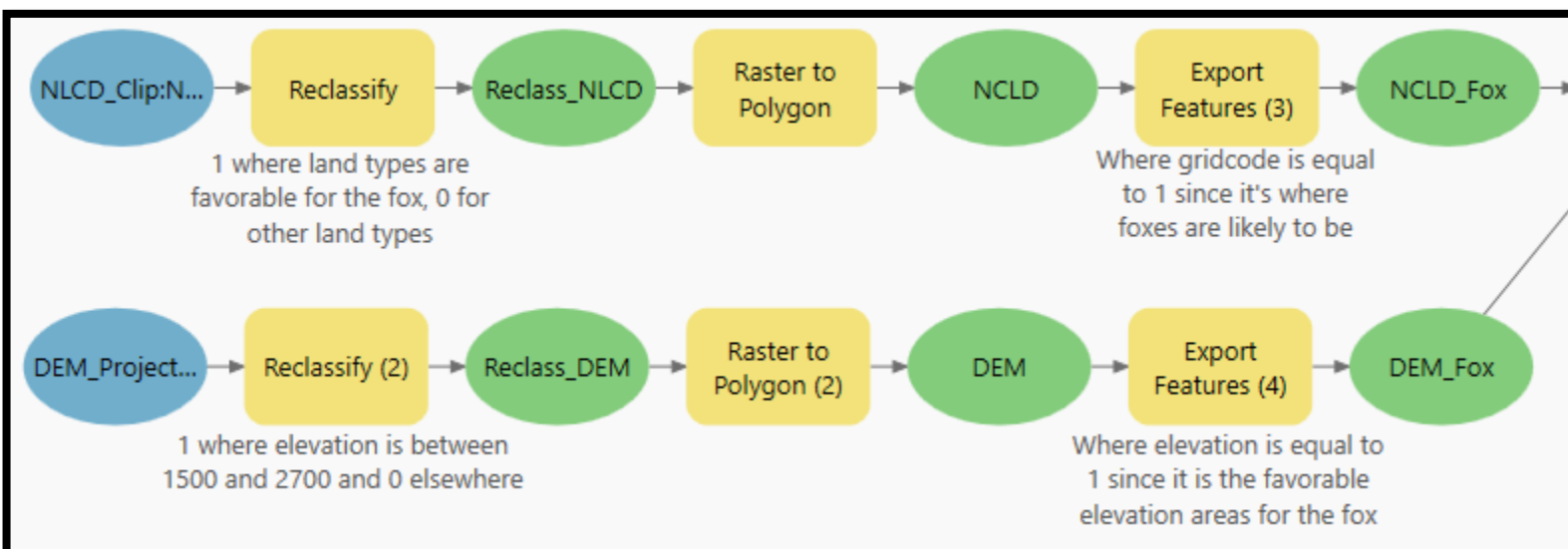


Figure 2 . NLCD and DEM processing steps to determine areas with favorable elevation and land cover types (zoomed in from Figure 1)

PART A — FIGURES (CONTINUED)

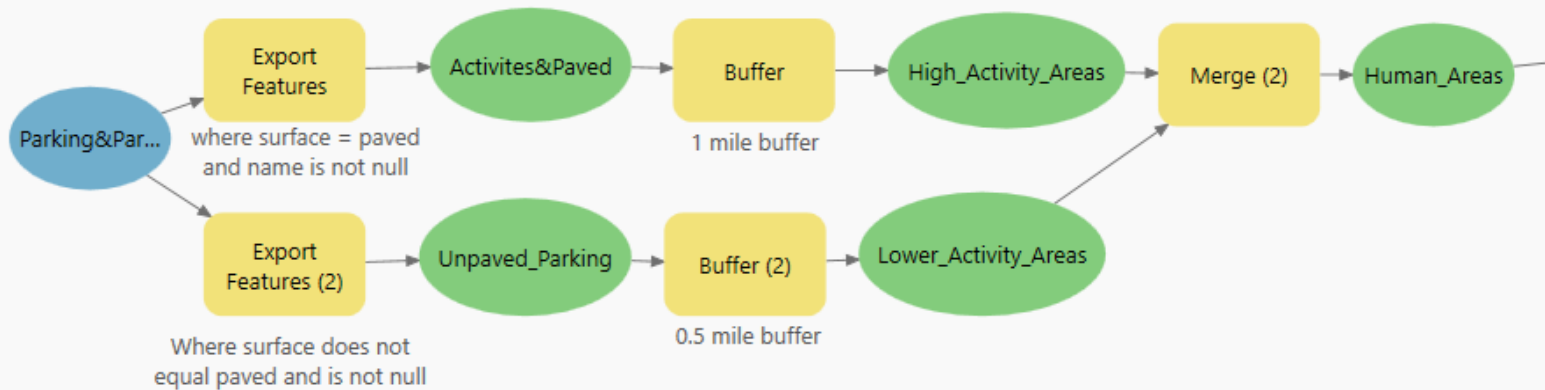


Figure 3 . Process for determining human zones given parking areas and activity points (zoomed in from Figure 1)

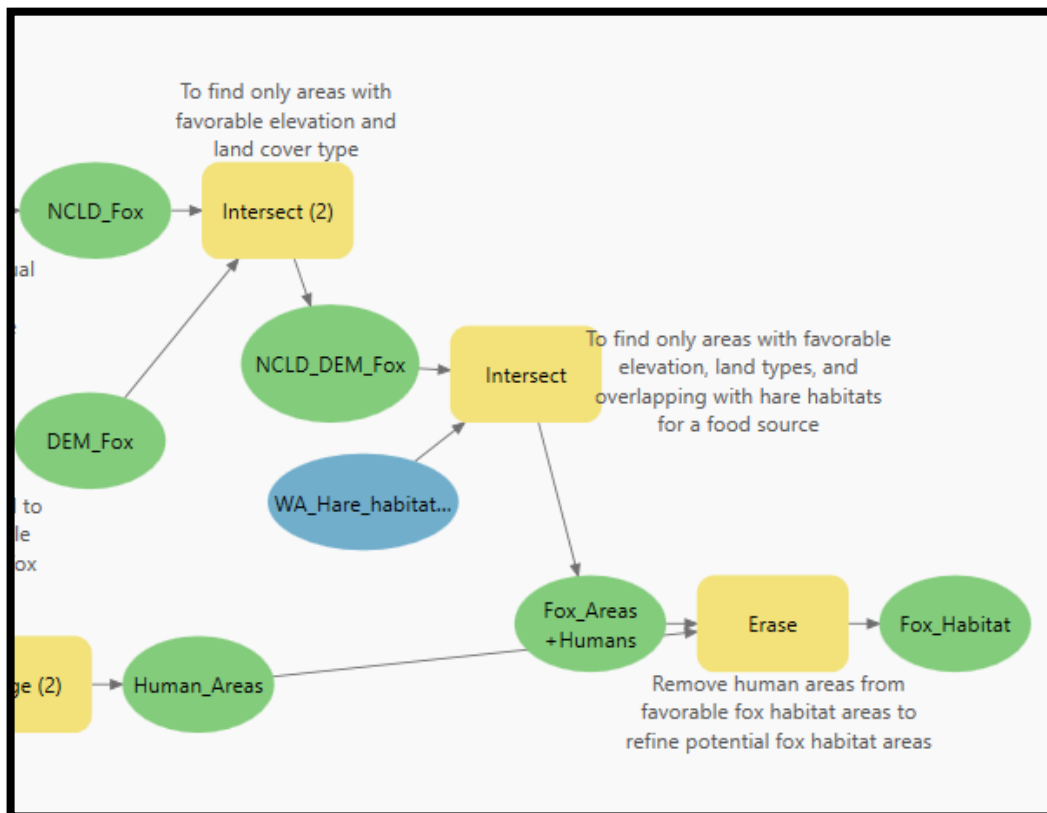


Figure 4 . Final steps for combining human areas and favorable land areas to determine the most likely fox habitat areas (zoomed in from Figure 1)

PART A (EXPLANATION - PART 1)

The first step (seen easily in figure 1) was to reclassify the raster layers to put a 1 where the conditions were favorable for foxes and a 0 where the conditions are not favorable for foxes. For land cover types, perennial ice/snow, barren land, deciduous forest, shrub/scrub, and grassland/herbaceous lands were considered favorable, so they were each assigned a 1 while all other types were assigned a zero. For elevation, a 1 was assigned to any elevation between 1500 and 2700 meters since the Cascade Red Fox is found in this elevation range and all other elevation were assigned a zero. Then each the NLCD and DEM raster layers were converted to polygons. Once they were polygons, the export features tool was used to create layers only where the grid code was equal to 1 and therefore favorable to the foxes. Both layers were then inputs for the intersect tool which created a layer that only included areas that included both a favorable land type and a favorable elevation (see figure 4). The intersect tool was then used a second time to further refine the results to only areas that had favorable elevation, land types, and overlapped with snow hare habitats since they are a common food source of the Cascade Red Fox.

Moving onto determining human areas (see figure 3) which the foxes are likely to avoid, the export features tool was used first to create two new layers. The first new layer includes all the park activity points and paved parking locations while the second layer only includes non-paved parking areas such as gravel lots. Then the buffer tool was used to create a circle around each point since it wouldn't be good for the foxes to be too close to humans

PART A (EXPLANATION - PART 2)

The buffer distance around the paved and activity areas is one mile since these areas are busier, and the distance around the unpaved areas is a half-mile since they aren't as likely to be crowded. The two buffer layers were merged into one layer of human areas to make the next step simpler.

Finally, to determine the best habitat areas for the Cascade Red Fox (see figure 4), the erase tool was used to subtract the human areas from the possible fox area layer described in the first paragraph to produce a final layer that shows the most favorable areas for the Cascade Red Fox in Washington State. This achieves the goal of the spatial analysis which was to build a suitability analysis for the Cascade Red Fox to find out where it is most likely to live given land cover types, elevation, distance from humans, and intersection with snow hare habitat for a food source. The final layer includes all the places that are the most favorable (such as a good food source) and leaves out areas that are unfavorable (such as near humans).

The major change to the model between lab 8 and the final report was to add labels to explain what is occurring in each of the processes and increase the font size for readability. Additionally, one of the steps was also corrected to use the intersect tool instead of the merge tool to find only the areas that the favorable conditions were both present. The end goal of the model was to determine suitable habitat locations for the Cascade Red Fox. In order to achieve this goal, we first had to determine the ideal land cover areas, elevations, overlap with snow hare habitats for food supply, and distance from human areas.

PART B (FIGURE)

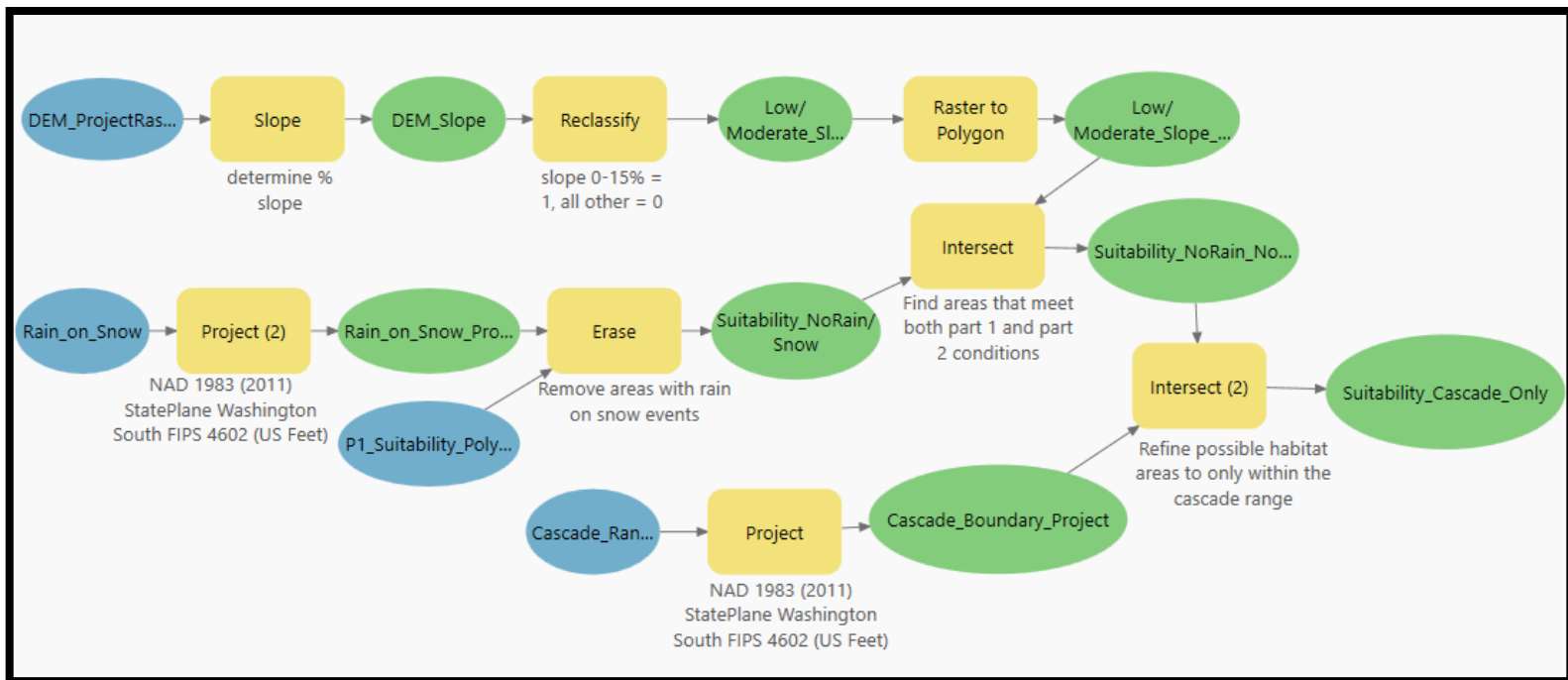


Figure 5 . Process for refining habitat locations to exclude areas with rain on snow events and areas with high slope, and areas outside of the Cascade range

PART B (EXPLANATION)

After determining suitable habitat areas for the Cascade Red Fox in the previous step, this step focuses on finding habitat areas with the highest suitability for the foxes (see Figure 5).

The first step was to use the elevation raster and run the slope tool to determine the percent slope for each area on the grid. Second, the reclassify tool was used to set areas with a slope between 0 and 15% to 1 and the rest of the areas with a higher slope to 0 since Cascade Red Foxes prefer low to moderate slope areas. Lastly, this raster file was converted to a polygon in preparation for the next steps.

The rain on snow file was projected to be in the same projection as the rest of the layers and then the erase tool was used to remove these areas from the suitable habitats layer created in the previous model since foxes since rain on snow events make it difficult for the foxes to hunt. The intersect tool was used next with the “suitable habitats minus rain and snow” layer to refine the areas to now also only include areas with low to moderate slopes.

The boundary for the Cascade mountain range was projected to match the coordinate system of the rest of the project and then the intersect tool was used with the Cascade range boundary and the high suitable habitats layer to refine the suitable habitats to only within the Cascade range since even though there are other areas that meet the criteria, the Cascade Red Foxes are only found within the Cascade mountain range.

The goal of this step of the analysis was to further refine the suitable habitats to make them more accurate to where the foxes may actually live. To do so, additional factors such as areas with rain on snow events and the slope of the land were added to the model.

PART C – LAYOUT 1

Washington State Public Lands

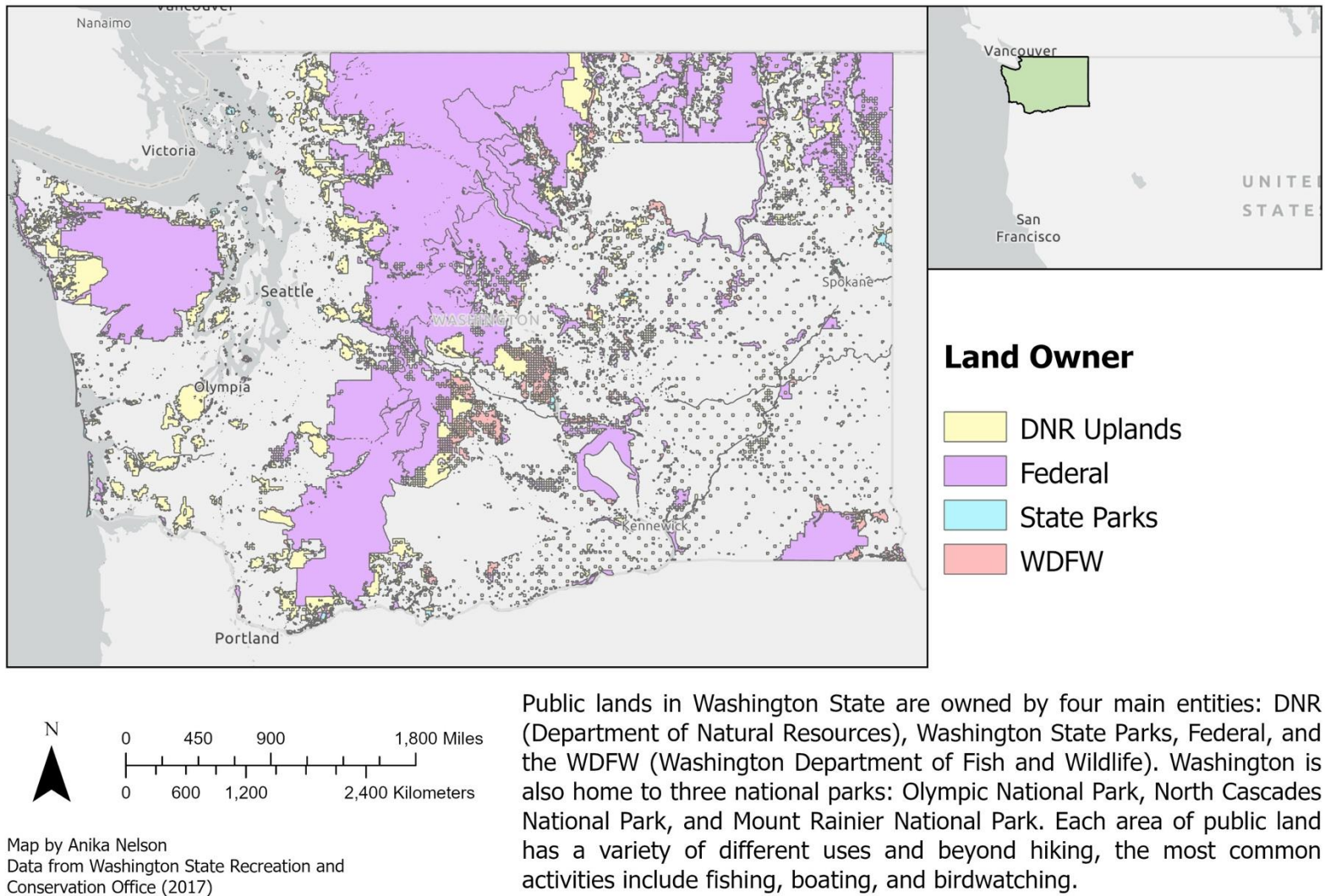


Figure 6 – Washington State Public Lands by owner.

Different public lands have different uses and levels of human activity which affect the habitats of wild animals, including the Cascade Red Fox. Additionally, each type of land is used differently. Some are primarily for recreation, such as State Parks which often have more infrastructure than the other areas such as DNR that have less recreational infrastructure but may allow industry, such as logging. The map was also updated to make the locator map more intuitive with an outline of Washington State.

PART C – LAYOUT 2

Factors Used to Determine Suitable Habitat Locations for the Cascade Red Fox

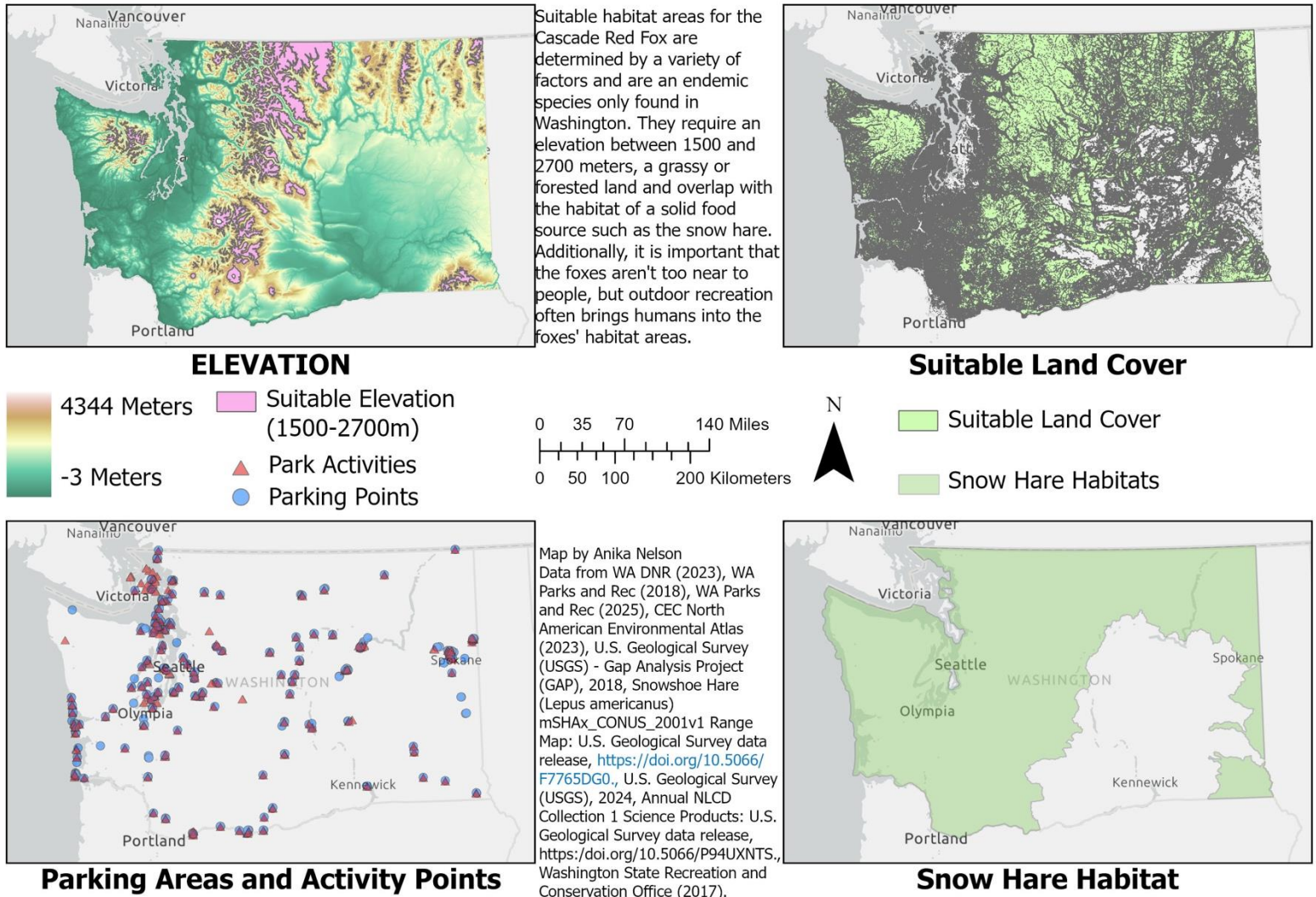


Figure 7 – Factors Used to Determine Suitable Habitat Locations for the Cascade Red Fox.

Factors include elevation, land cover type, snow hare (prey) habitat, and proximity to human areas. Finding an area that meets all the favorable factors leads to a small habitat area for the fox which is increasingly threatened by human-caused climate change. The rough estimate from the four maps shows that the most suitable habitat is likely within the Cascade mountain range.

PART C – LAYOUT 3

Suitable Locations for Cascade Red Fox Habitats

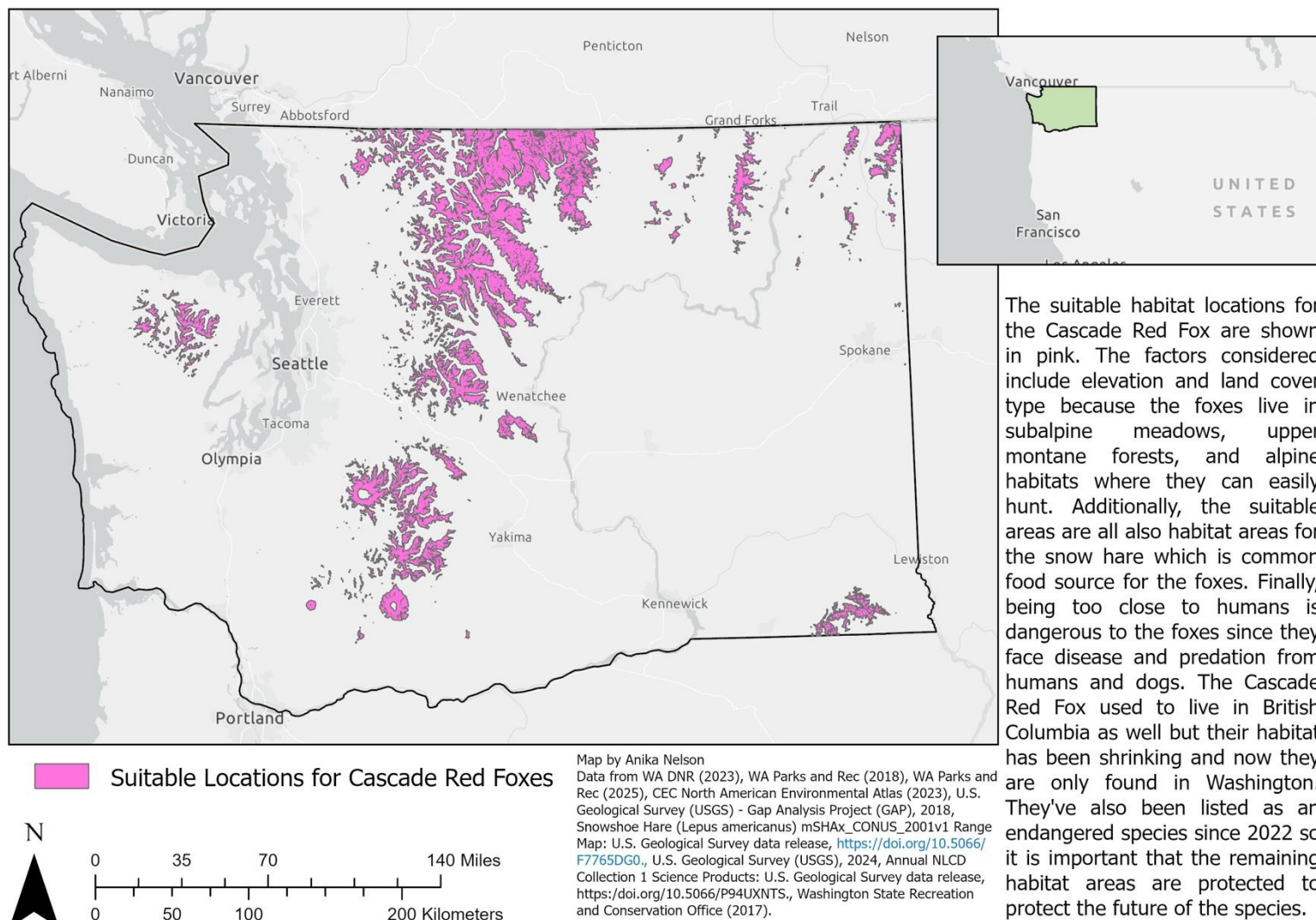


Figure 8 – Suitable Locations for Cascade Red Fox Habitats.

The Cascade Red Fox has specific habitat needs such as the right type of land cover and availability of prey to be able to hunt. It is important to protect their habitat since they are only found in Washington State and without access to proper habitat the already endangered foxes would likely go extinct in the wild. The largest continuous habitat areas are found in the northern areas of Cascade mountain range.

PART C – LAYOUT 4

Ideal Cascade Red Fox Habitat Locations

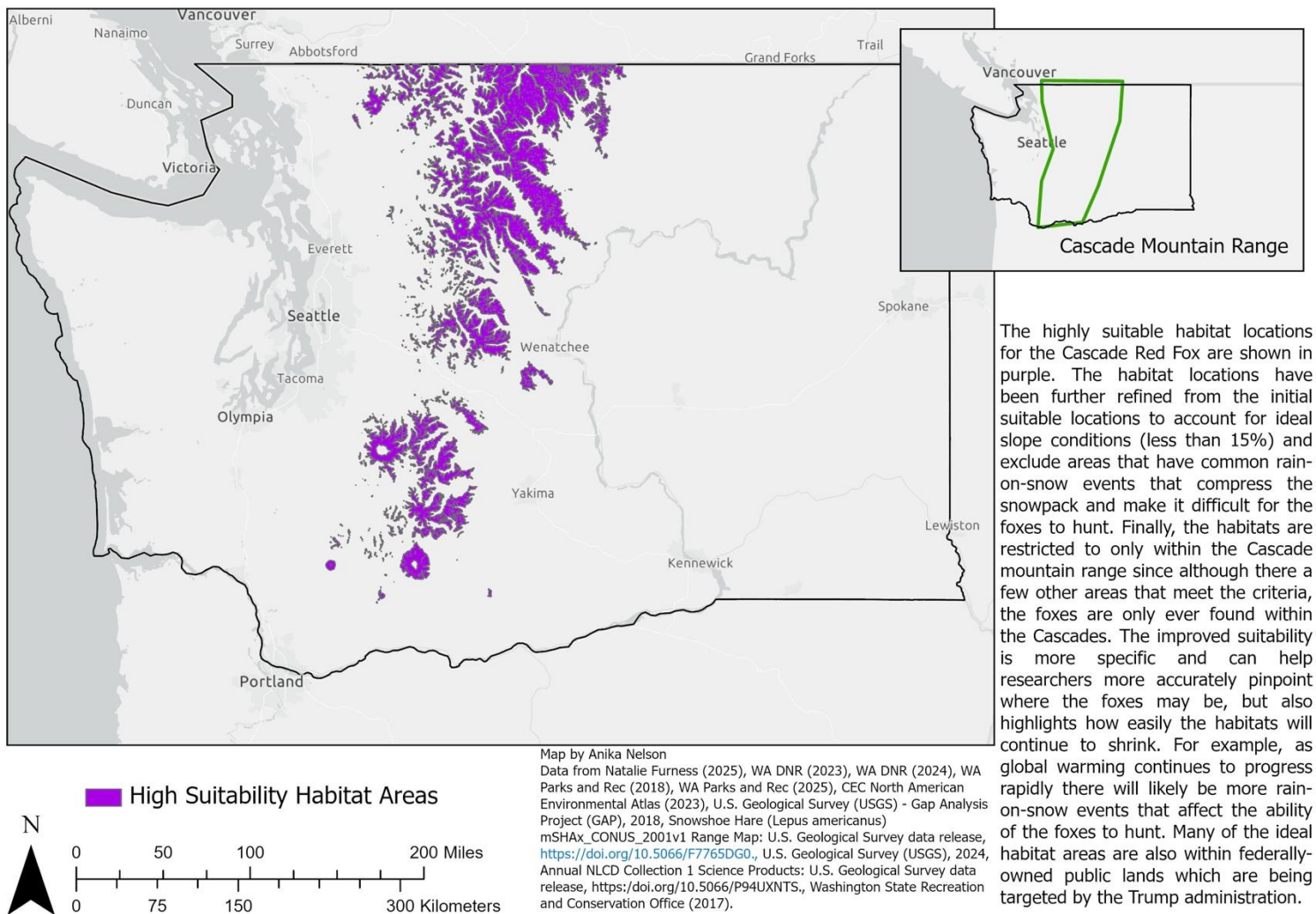


Figure 9 – Ideal Cascade Red Fox Habitat Locations

The most suitable habitat areas are still within the northern areas of the Cascade mountain range. It is important to note that the Cascade Red Foxes have already been pushed out of British Columbia due to habitat loss, so it is important to protect these habitat areas to ensure survival of the species. The prevalence of rain-on-snow events greatly reduced the suitable habitat areas and these events are likely to increase as global warming continues to alter weather patterns.

PART C

OneDrive > Anika - Western Washington University > ENVS 320 > FinalProject

Name	Status	Date modified	Type	Size
.backups	✓	7/21/2025 3:53 PM	File folder	
Data	✓	7/21/2025 2:37 PM	File folder	
FinalProject.gdb	✓	7/21/2025 3:53 PM	File folder	
GpMessages	✓	7/21/2025 2:46 PM	File folder	
ImportLog	✓	7/21/2025 2:38 PM	File folder	
Index	✓	7/21/2025 3:27 PM	File folder	
FinalProject	✓	7/21/2025 3:47 PM	ArcGIS Project File	35 KB
FinalProject.atbx	✓	7/21/2025 3:43 PM	ATBX File	36 KB
Park_Activities.cpg	✓	7/21/2025 3:04 PM	CPG File	1 KB
Park_Activities.dbf	✓	7/21/2025 3:04 PM	DBF File	164 KB
Park_Activities.prj	✓	7/21/2025 3:04 PM	PRJ File	1 KB
Park_Activities.sbn	✓	7/21/2025 3:04 PM	SBN File	9 KB
Park_Activities.sbx	✓	7/21/2025 3:04 PM	SBX File	1 KB
Park_Activities.shp	✓	7/21/2025 3:04 PM	SHP File	25 KB
Park_Activities.shp	✓	7/21/2025 3:04 PM	Microsoft Edge H...	9 KB
Park_Activities.shx	✓	7/21/2025 3:04 PM	SHX File	8 KB
Park_Activities_1.cpg	✓	7/21/2025 3:05 PM	CPG File	1 KB
Park_Activities_1.dbf	✓	7/21/2025 3:05 PM	DBF File	164 KB
Park_Activities_1.prj	✓	7/21/2025 3:05 PM	PRJ File	1 KB
Park_Activities_1.sbn	✓	7/21/2025 3:05 PM	SBN File	9 KB
Park_Activities_1.sbx	✓	7/21/2025 3:05 PM	SBX File	1 KB
Park_Activities_1.shp	✓	7/21/2025 3:05 PM	SHP File	25 KB

OneDrive > Anika - Western Washington University > ENVS 320 > FinalProject > Data

Figure 10 –
Contents inside
Final Project
Folder

OneDrive > Anika - Western Washington University > ENVS 320 > FinalProject > Data

Name	Status	Date modified	Type	Size
a0000000a.EditingTemplateRelsByDestina...	✓	7/21/2025 2:37 PM	ATX File	5 KB
a0000000a.EditingTemplateRelsByOriginl...	✓	7/21/2025 2:37 PM	ATX File	5 KB
a0000000a.EditingTemplateRelsByType.atx	✓	7/21/2025 2:37 PM	ATX File	5 KB
a0000000a.gdbindexes	✓	7/21/2025 2:37 PM	GDBINDEXES File	1 KB
a0000000a.gdbtable	✓	7/21/2025 2:37 PM	GDBTABLE File	1 KB
a0000000a.gdbtablx	✓	7/21/2025 2:37 PM	GDBTABLX File	1 KB
a0000000b.gdbindexes	✓	7/21/2025 2:37 PM	GDBINDEXES File	1 KB
a0000000b.gdbtable	✓	7/21/2025 2:37 PM	GDBTABLE File	2,247 KB
a0000000b.gdbtablx	✓	7/21/2025 2:37 PM	GDBTABLX File	6 KB
a0000000b.horizon	✓	7/21/2025 2:37 PM	HORIZON File	1 KB
a0000000b	✓	7/21/2025 2:37 PM	SPX Audio File (VL...	5 KB

Figure 11 – Contents inside Data subfolder

PART D - DISCUSSION

The ideal habitat locations are far smaller than the suitable habitat conditions. One reason for the decreased area is due to the high prevalence of rain-on-snow conditions within Washington state that compress the snow and make it hard for the foxes to hunt, so in an ideal location, there would be very few of these events to ensure that their food supply is consistent. As the climate continues to warm, the prevalence of rain-on-snow events within the alpine areas the foxes call home are likely to increase. The ideal locations also have a low slope since the foxes like to hunt in meadow areas rather than mountainsides. Additionally, some of the suitable locations fall outside of the Cascade mountain range, but the foxes have only ever been found within the Cascade range, so it is important to only consider suitable habitat locations within the Cascade range. For both the ideal habitat locations and suitable habitat locations, the largest continuous areas of habitat are within the northern Cascade range, just south of the Canadian border.

Running the refined suitability analysis narrows the result which is helpful if researchers are only interested in the most likely areas to find the Cascade red foxes in real life rather than all the possible areas that could support them. The refined suitability analysis is beneficial since it can help researchers pinpoint the areas of high priority for habitat conservation. Additionally, if researchers are looking to set cameras or traps for the foxes, having a reduced area with the highest likelihood of encountering a fox will increase their chances for success.

REFERENCES

CEC North American Environmental Atlas (2023)

Natalie Furness (2025)

The Cascade Red Fox's fate in Washington State. (n.d.). National Wildlife Federation. <https://www.nwf.org/Magazines/National-Wildlife/2023/Fall/Conservation/Cascade-Red-Fox>

U.S. Geological Survey (USGS) - Gap Analysis Project (GAP), 2018, Snowshoe Hare (*Lepus americanus*) mSNHax_CONUS_2001v1 Range Map: U.S. Geological Survey data release, <https://doi.org/10.5066/F7765DGO>.

U.S. Geological Survey (USGS), 2024, Annual NLCD Collection 1 Science Products: U.S. Geological Survey data release, <https://doi.org/10.5066/P94UXNTS>

WA DNR (2023)

WA DNR (2024)

WA Parks and Rec (2018)

WA Parks and Rec (2025)

Washington State Recreation and Conservation Office (2017)